



DESARROLLO DE SOFTWARE PARA PLATAFORMAS MOVILES

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Memories

- ✓ What does useEffect do with []?
- ✓ What happens if I modify the state directly?
- ✓ Why do we use key?
- ✓ How does a child execute something from its parent?



Memories

How many times is this effect executed?

```
const [count, setCount] = useState(0);  
  
useEffect(() => {  
  console.log("Hola");  
}, [ ]);
```



Memories

What happens if I modify the state directly?

Is this the same?

```
const [contacts, setContacts] = useState([]);  
  
const addContact = () => {  
  contacts.push({ name: "Nuevo" });  
};
```

```
const [contacts, setContacts] = useState([]);  
  
const addContact = () => {  
  setContacts([...contacts, { name: "Nuevo" }]);  
};
```



Memories

✓ Why do we use key?

```
{contacts.map((c, index) => (  
  <div key={index}>{c.name}</div>  
))  
}}
```



Memories

✓ How does a child execute something from its parent?

```
const deleteContact = (id) => {  
  setContacts(contacts.filter(c => c.id !== id));  
  
  <ContactItem  
    contact={contact}  
    onDelete={deleteContact}  
  />
```

```
const ContactItem = ({ contact, onDelete }) => {  
  return (  
    <button onClick={() => onDelete(contact.id)}>  
      Eliminar  
    </button>  
  );  
};
```



Ionic

- ✓ An open-source, cross-platform framework that allows to create high-performance native mobile apps, progressive web apps (PWAs), and desktop applications.
- ✓ Use standard web technologies such as HTML, CSS, and JavaScript.
- ✓ Integrated with popular frameworks like Angular, React, and Vue, facilitating hybrid development.



Installing Ionic

`npm i -g @ionic/cli`

<https://ionicframework.com/docs/intro/cli>



Creating a new Ionic Project

ionic start **contactos**

Pick a framework!

Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the `--type` option.

```
? Framework: (Use arrow keys)
> Angular | https://angular.io
  React   | https://reactjs.org
  Vue     | https://vuejs.org
```

Let's pick the perfect starter template!

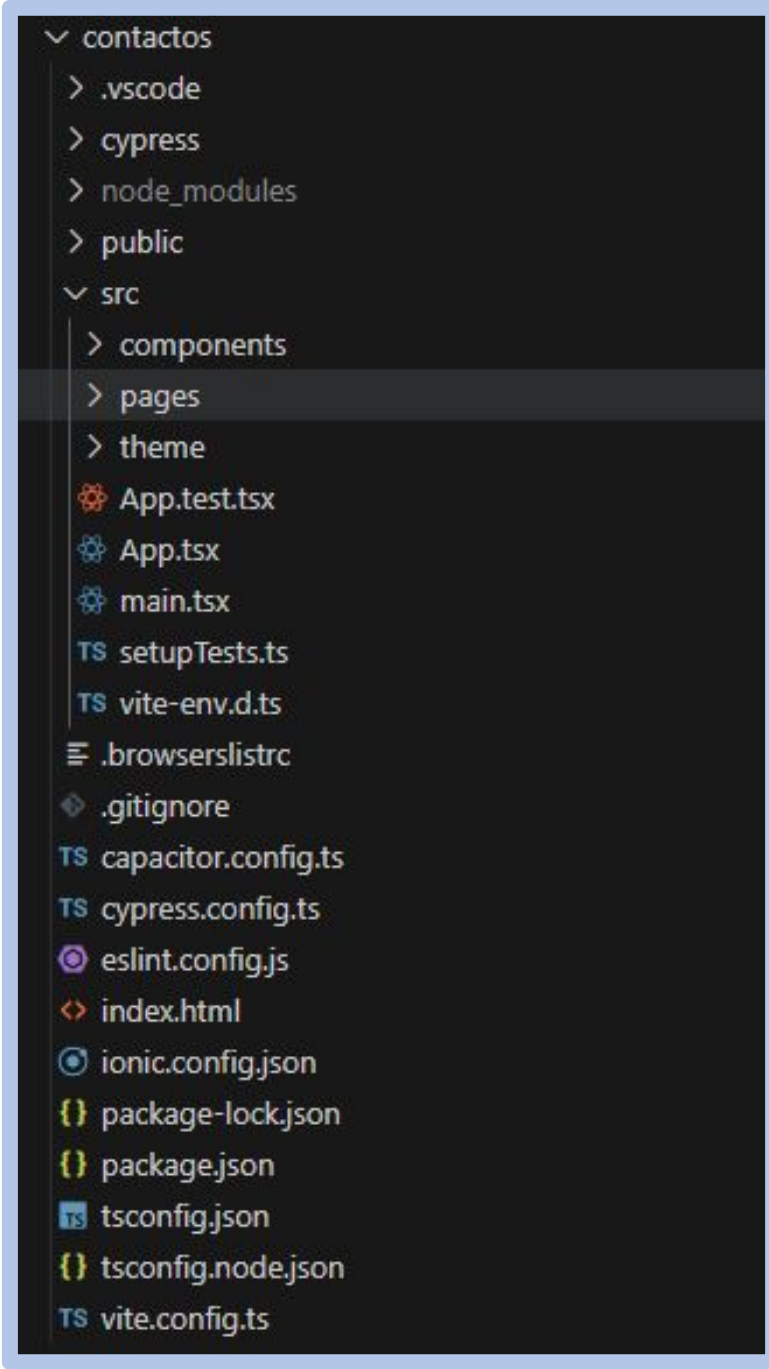
Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this prompt next time, supply `template`, the second argument to `ionic start`.

```
? Starter template: (Use arrow keys)
> blank      | A blank starter project
  list       | A starting project with a list
  my-first-app | A template for the "Build Your First App" tutorial
  sidemenu   | A starting project with a side menu with navigation in the content area
  tabs       | A starting project with a simple tabbed interface
```

...

cd **contactos**
ionic serve

Folder Structure



```

  contactos
  ├── .vscode
  ├── cypress
  ├── node_modules
  ├── public
  └── src
      ├── components
      ├── pages
      ├── theme
      ├── App.test.tsx
      ├── App.tsx
      ├── main.tsx
      ├── setupTests.ts
      ├── vite-env.d.ts
      ├── .browserslistrc
      ├── .gitignore
      ├── capacitor.config.ts
      ├── cypress.config.ts
      ├── eslint.config.js
      ├── index.html
      ├── ionic.config.json
      ├── package-lock.json
      ├── package.json
      ├── tsconfig.json
      ├── tsconfig.node.json
      └── vite.config.ts

```

A screenshot of a file explorer interface showing a project folder structure. The 'src' folder is expanded, revealing subfolders 'components', 'pages', and 'theme', along with various TypeScript and configuration files. The 'pages' folder is currently selected and highlighted.

Basic Anatomy of Ionic

- ✓ **IonApp**
- ✓ IonPage
- ✓ IonHeader
- ✓ IonToolbar
- ✓ IonTitle
- ✓ IonContent
- ✓ IonButton
- ✓ IonList
- ✓ IonItem

```
App.tsx
contactos > src > App.tsx > ...
27  /*
28
29  /* import '@ionic/react/css/palettes/dark.always.css'; */
30  /* import '@ionic/react/css/palettes/dark.class.css'; */
31  import '@ionic/react/css/palettes/dark.system.css';
32
33  /* Theme variables */
34  import './theme/variables.css';
35
36  setupIonicReact();
37
38  const App: React.FC = () => (
39    <IonApp>
40      <IonReactRouter>
41        <IonRouterOutlet>
42          <Route exact path="/home">
43            <Home />
44          </Route>
45          <Route exact path="/">
46            <Redirect to="/home" />
47          </Route>
48        </IonRouterOutlet>
49      </IonReactRouter>
50    </IonApp>
51  );
52
53  export default App;
54
```

Basic Anatomy of Ionic

-

Page Structure

- ✓ IonApp
- ✓ IonPage
- ✓ IonHeader
- ✓ IonToolbar
- ✓ IonTitle
- ✓ IonContent
- ✓ IonButton
- ✓ IonList
- ✓ IonItem

```
Home.tsx X
contactos > src > pages > Home.tsx > ...
1  import { IonContent, IonHeader, IonPage, IonTitle, IonToolbar } from '@ionic/react';
2  import ExploreContainer from '../components/ExploreContainer';
3  import './Home.css';
4
5  const Home: React.FC = () => {
6    return (
7      <IonPage>
8        <IonHeader>
9          <IonToolbar>
10           <IonTitle>Blank</IonTitle>
11         </IonToolbar>
12       </IonHeader>
13       <IonContent fullscreen>
14         <IonHeader collapse="condense">
15           <IonToolbar>
16             <IonTitle size="large">Blank</IonTitle>
17           </IonToolbar>
18         </IonHeader>
19         <ExploreContainer />
20       </IonContent>
21     </IonPage>
22   );
23 };
24
25 export default Home;
26
```

Basic Anatomy of Ionic - Buttons

- ✓ IonApp
- ✓ IonPage
- ✓ IonHeader
- ✓ IonToolbar
- ✓ IonTitle
- ✓ IonContent
- ✓ **IonButton**
- ✓ IonList
- ✓ IonItem

```
const ExploreContainer: React.FC<ContainerProps> = () => {  
  return (  
    <div id="container">  
      <button> Esto es un botón </button>  
      <ul>  
        <li> Esto es una lista </li>  
        <li> Esto es una lista </li>  
        <li> Esto es una lista </li>  
      </ul>  
    </div>  
  );  
};  
  
export default ExploreContainer;
```

```
const ExploreContainer: React.FC<ContainerProps> = () => {  
  return (  
    <div id="container">  
      <IonButton> Esto es un botón </IonButton>  
      <IonList>  
        <IonItem> Esto es una lista </IonItem>  
        <IonItem> Esto es una lista </IonItem>  
        <IonItem> Esto es una lista </IonItem>  
      </IonList>  
    </div>  
  );  
};  
  
export default ExploreContainer;
```

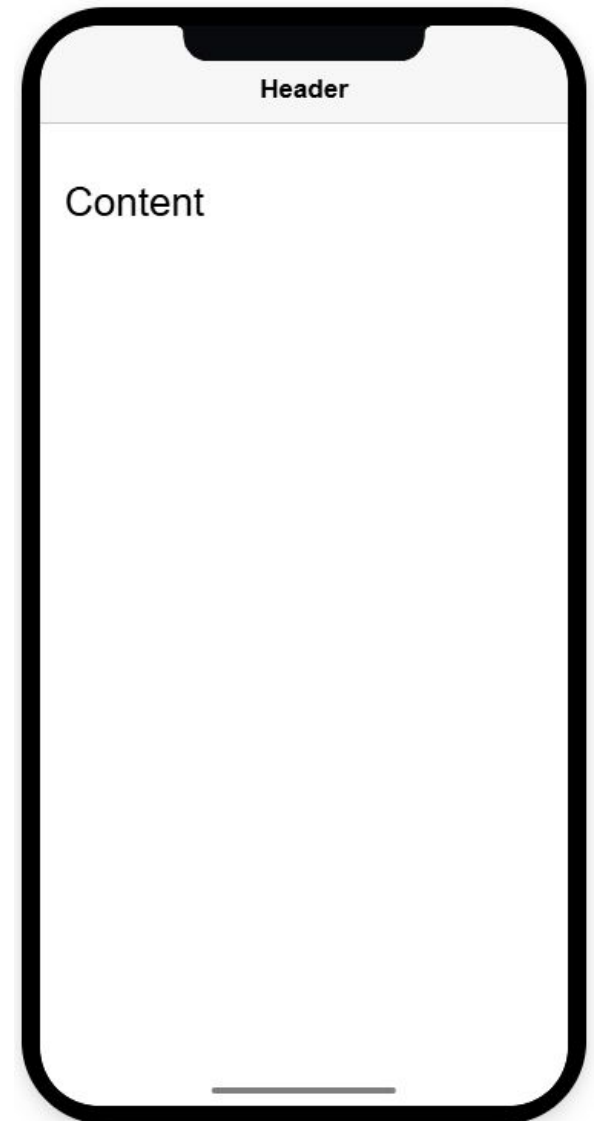
Header Layout

The simplest layout available consists of a header and content.

Most pages in an app generally have both of these, but a header is not required in order to use content.

```
<ionPage>
  <ionHeader>
    <ionToolbar>
      <ionTitle>Header</ionTitle>
    </ionToolbar>
  </ionHeader>

  <ionContent className="ion-padding">
    <h1>Content</h1>
  </ionContent>
</ionPage>
```

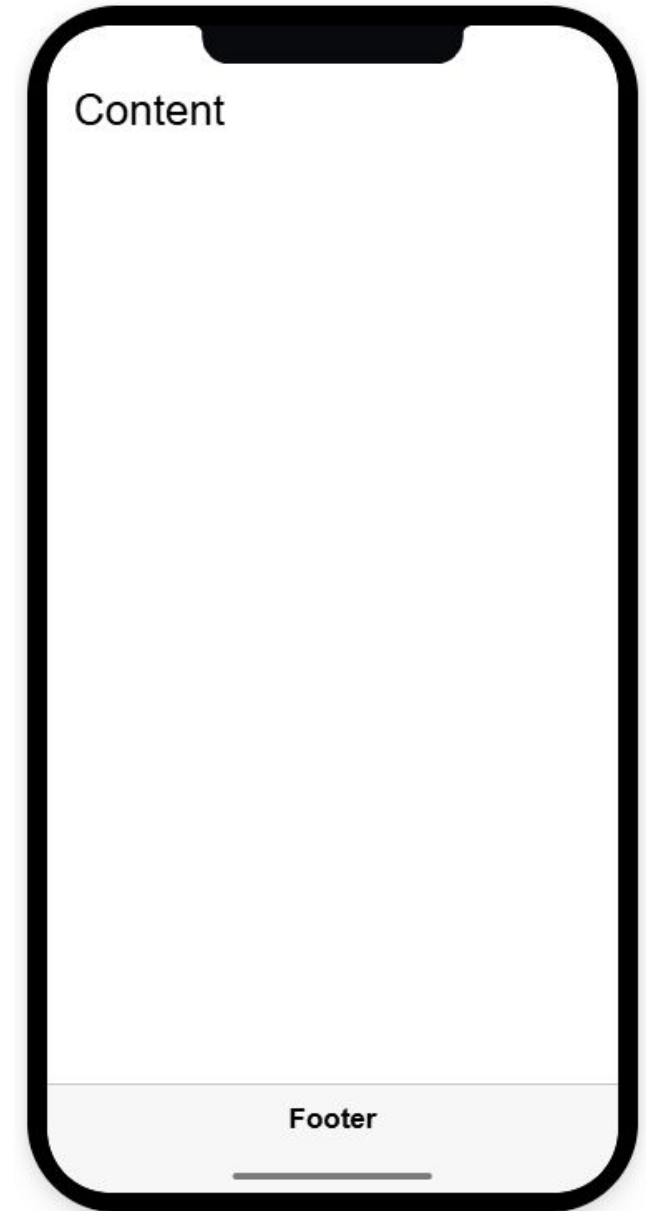


Footer Layout

While a toolbar in a header appears above the content, a footer appears below the content. A header and a footer can also be used together on the same page.

```
<ionPage>
  <ionContent className="ion-padding">
    <h1>Content</h1>
  </ionContent>

  <ionFooter>
    <ionToolbar>
      <ionTitle>Footer</ionTitle>
    </ionToolbar>
  </ionFooter>
</ionPage>
```



Tabs Layout

A layout consisting of horizontal tabs can be used to let the user quickly change between content views.

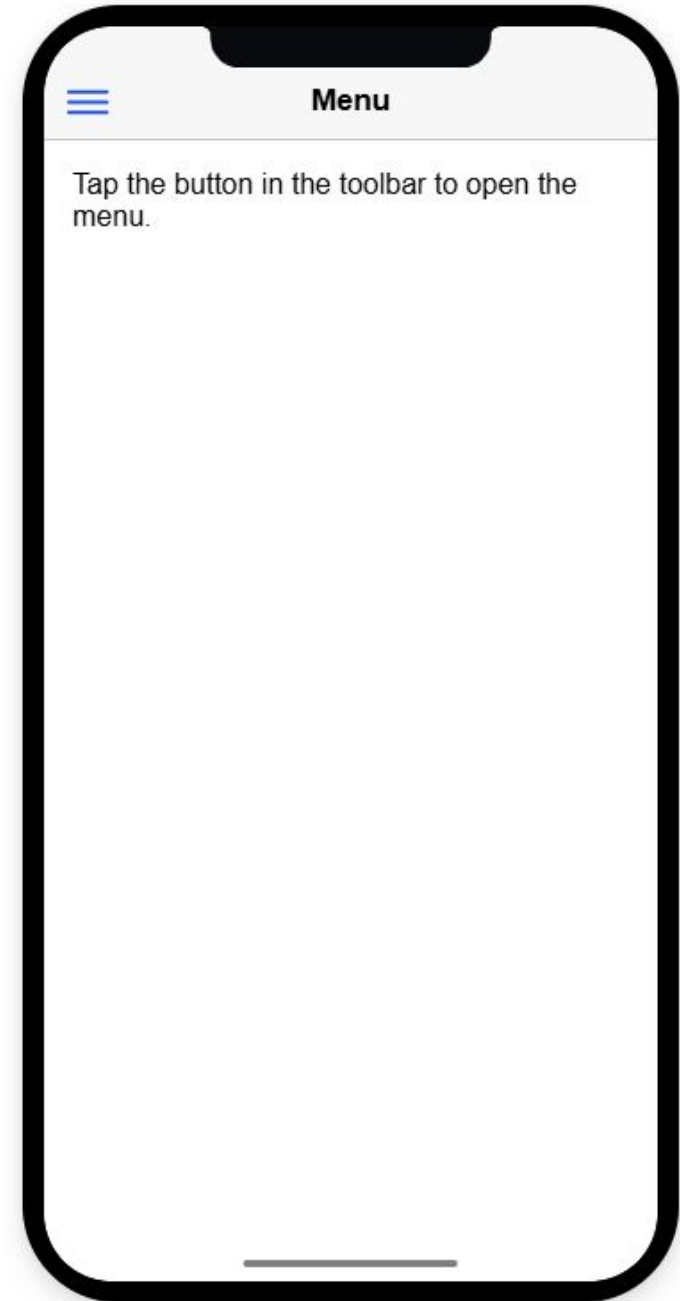
Each tab can contain static content or a navigation stack by using a router outlet or nav.



Menu Layout

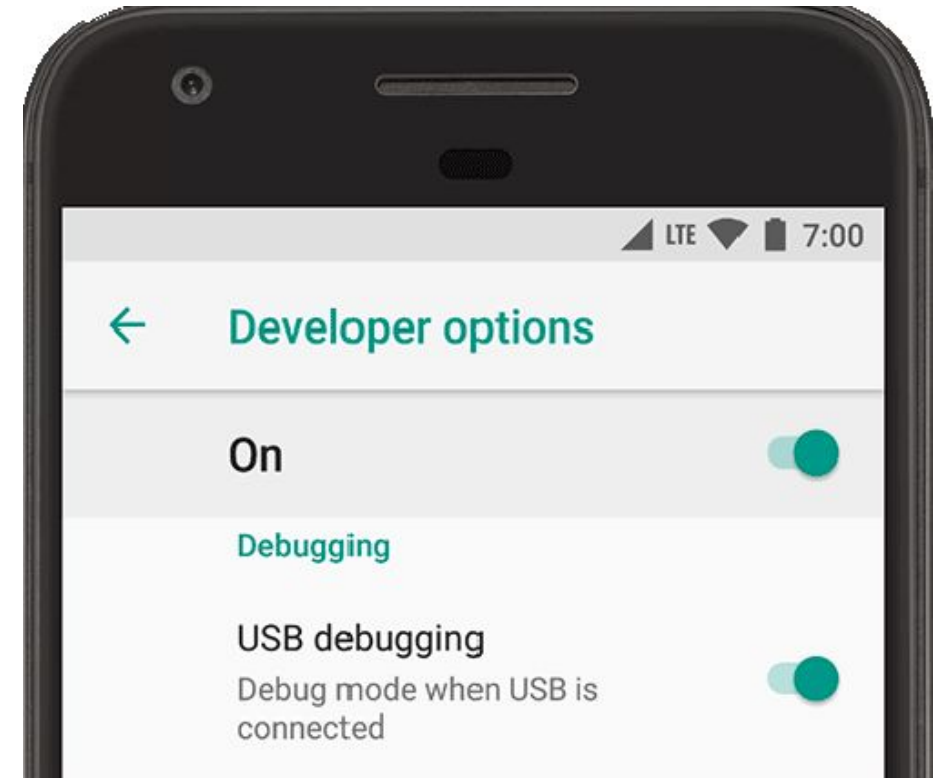
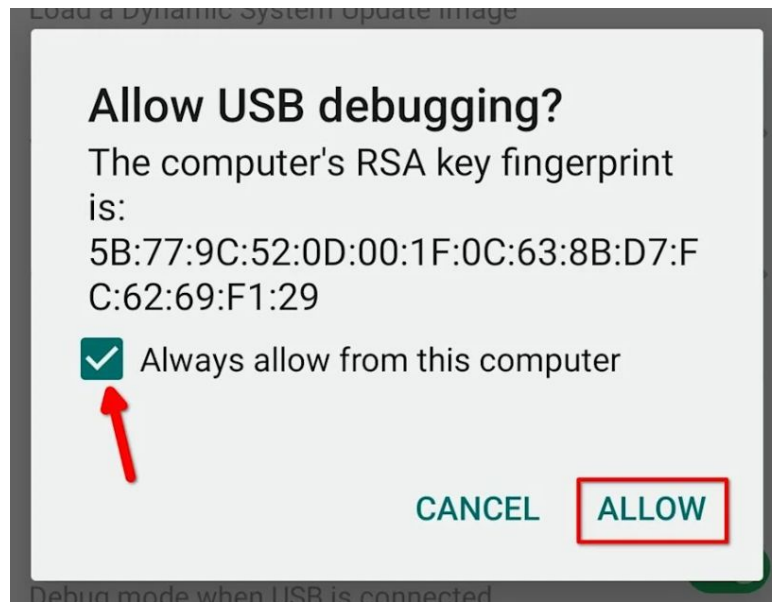
A standard layout among mobile apps includes the ability to toggle a side menu by clicking a button or swiping it open from the side.

Side menus are generally used for navigation, but they can contain any content.



Install into Android

- ✓ Must have Android Studio installed and configured
- ✓ Developer mode on your phone must be enabled.
 - ✓ Go to Settings
 - ✓ About phone
 - ✓ Tap "Build number" 7 times
 - ✓ Then, in developer options, Enable "USB debugging"
- ✓ After that, must connect your phone and allow USB Debugging

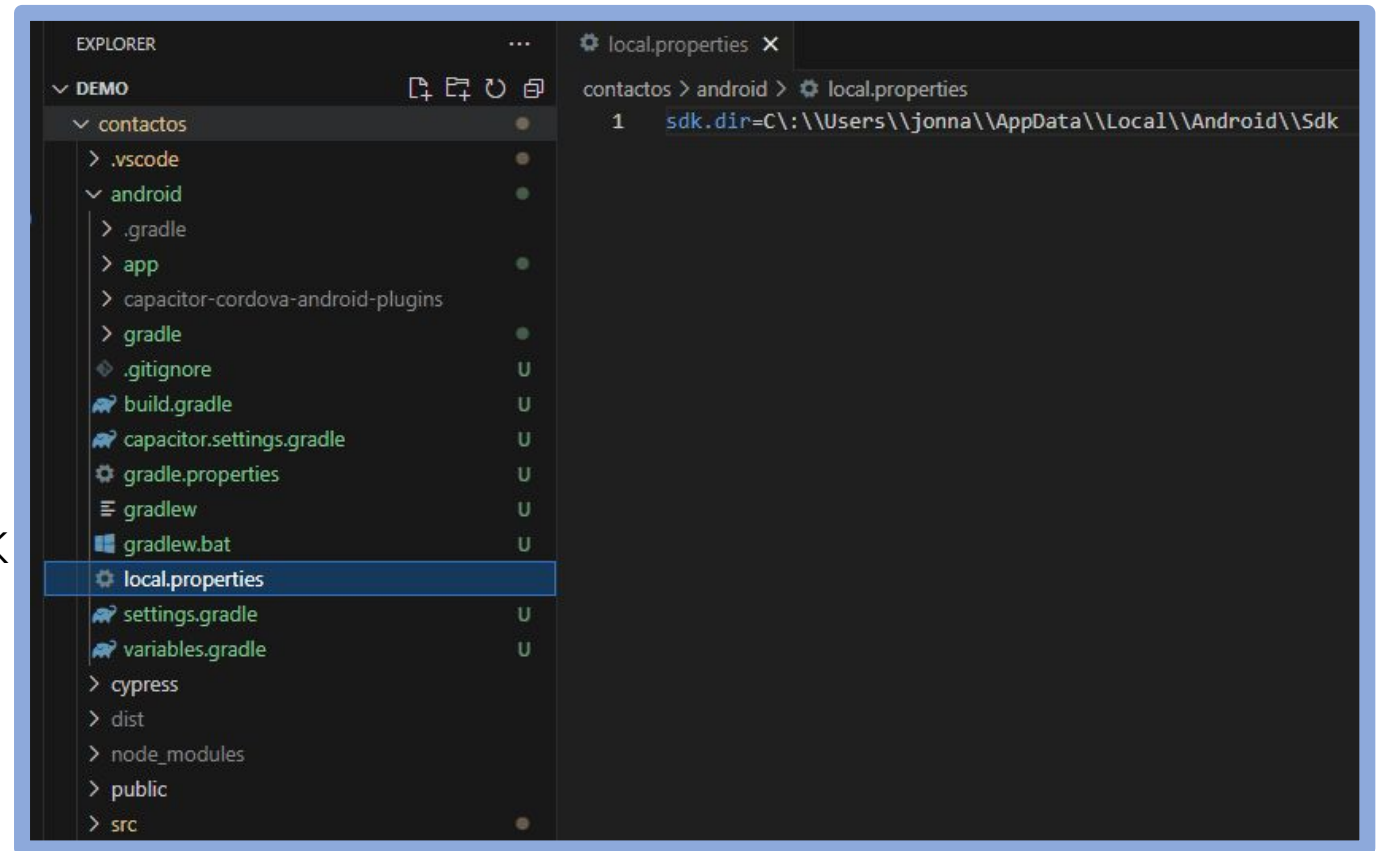


Install into Android

- ✓ Run this command in your visual studio project: **ionic cap run android**.

This command will run internally:

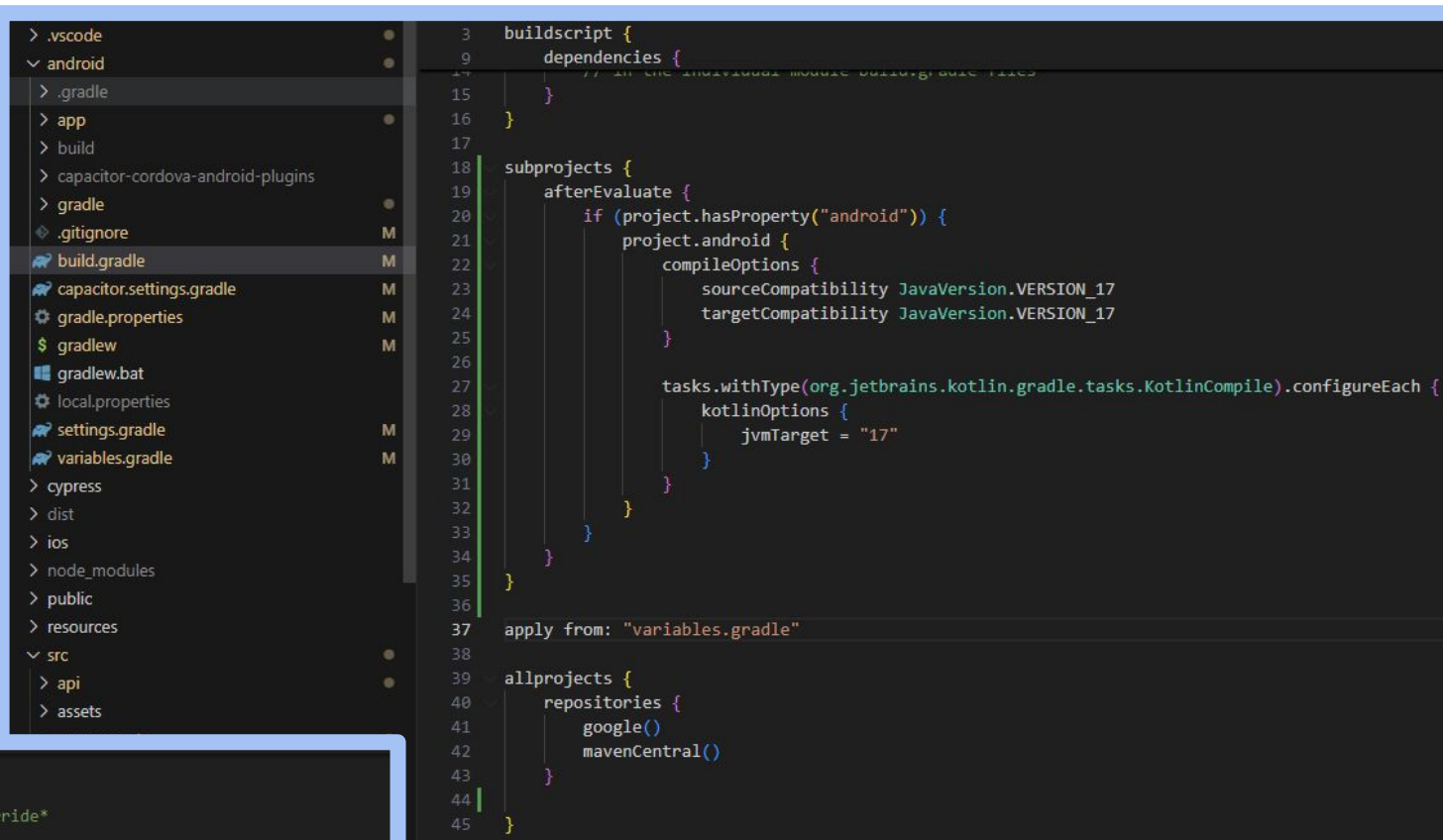
- ✓ ionic build: Builds the web app (React)
 - ✓ npx cap copy: Copies the files to the /android folder
 - ✓ npx cap sync: Syncs plugins
 - ✓ Open and run on Android (emulator or device)
- ✓ Could be necessary create the local.properties file to find the android SDK folder installed in your PC:



Fixing errors into Android

- ✓ Sometimes, could be necessary install and configure JAVA 17 in your project, **depending on** the Android Gradle settings on your PC:

<https://www.oracle.com/java/technologies/javase/jdk17-archive-downloads.html>

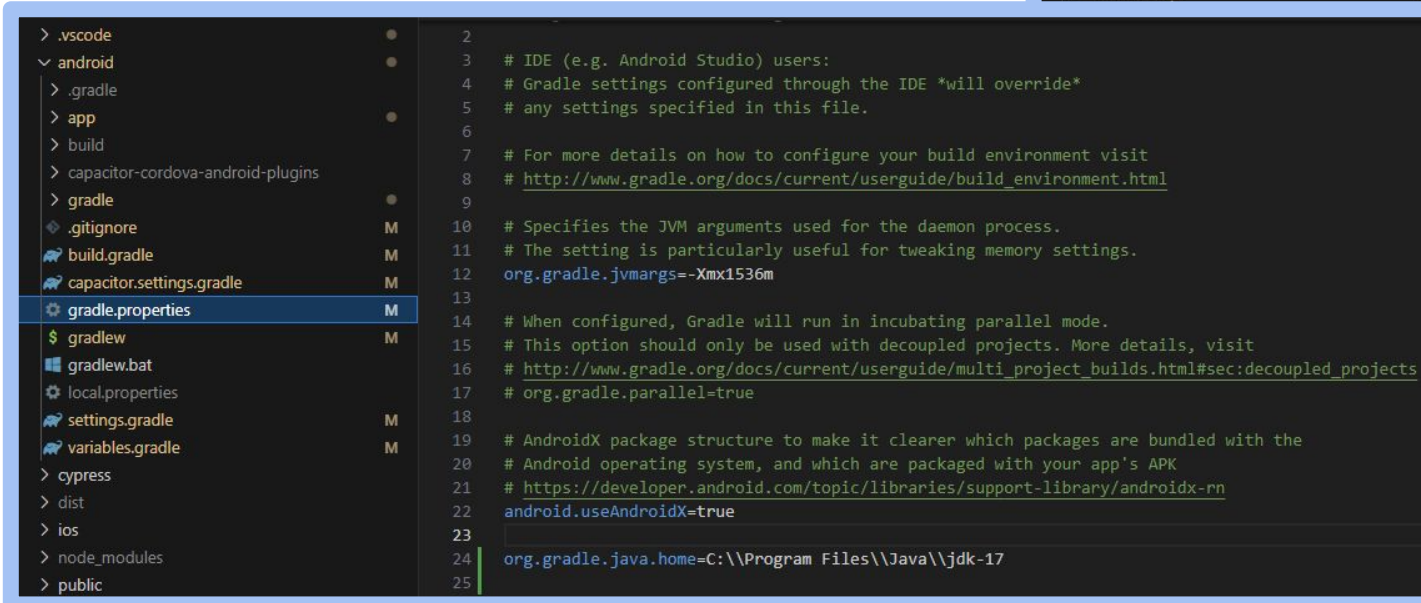


The screenshot shows the VS Code interface with the Android project file explorer on the left and the build.gradle file open in the editor. The file explorer shows the following structure:

- > .vscode
- ▼ android
 - > .gradle
 - > app
 - > build
 - > capacitor-cordova-android-plugins
 - > gradle
 - ◆ .gitignore
 - 🔗 build.gradle
 - 🔗 capacitor.settings.gradle
 - ⚙️ gradle.properties
 - 💰 gradlew
 - 📄 gradlew.bat
 - ⚙️ local.properties
 - 🔗 settings.gradle
 - 🔗 variables.gradle
 - > cypress
 - > dist
 - > ios
 - > node_modules
 - > public
 - > resources
 - ▼ src
 - > api
 - > assets

The build.gradle file contains the following code:

```
3 buildscript {
9     dependencies {
15     }
16 }
17
18 subprojects {
19     afterEvaluate {
20         if (project.hasProperty("android")) {
21             project.android {
22                 compileOptions {
23                     sourceCompatibility JavaVersion.VERSION_17
24                     targetCompatibility JavaVersion.VERSION_17
25                 }
26
27                 tasks.withType(org.jetbrains.kotlin.gradle.tasks.KotlinCompile).configureEach {
28                     kotlinOptions {
29                         jvmTarget = "17"
30                     }
31                 }
32             }
33         }
34     }
35 }
36
37 apply from: "variables.gradle"
38
39 allprojects {
40     repositories {
41         google()
42         mavenCentral()
43     }
44 }
45 }
```



The screenshot shows the VS Code interface with the Android project file explorer on the left and the gradle.properties file open in the editor. The file explorer shows the following structure:

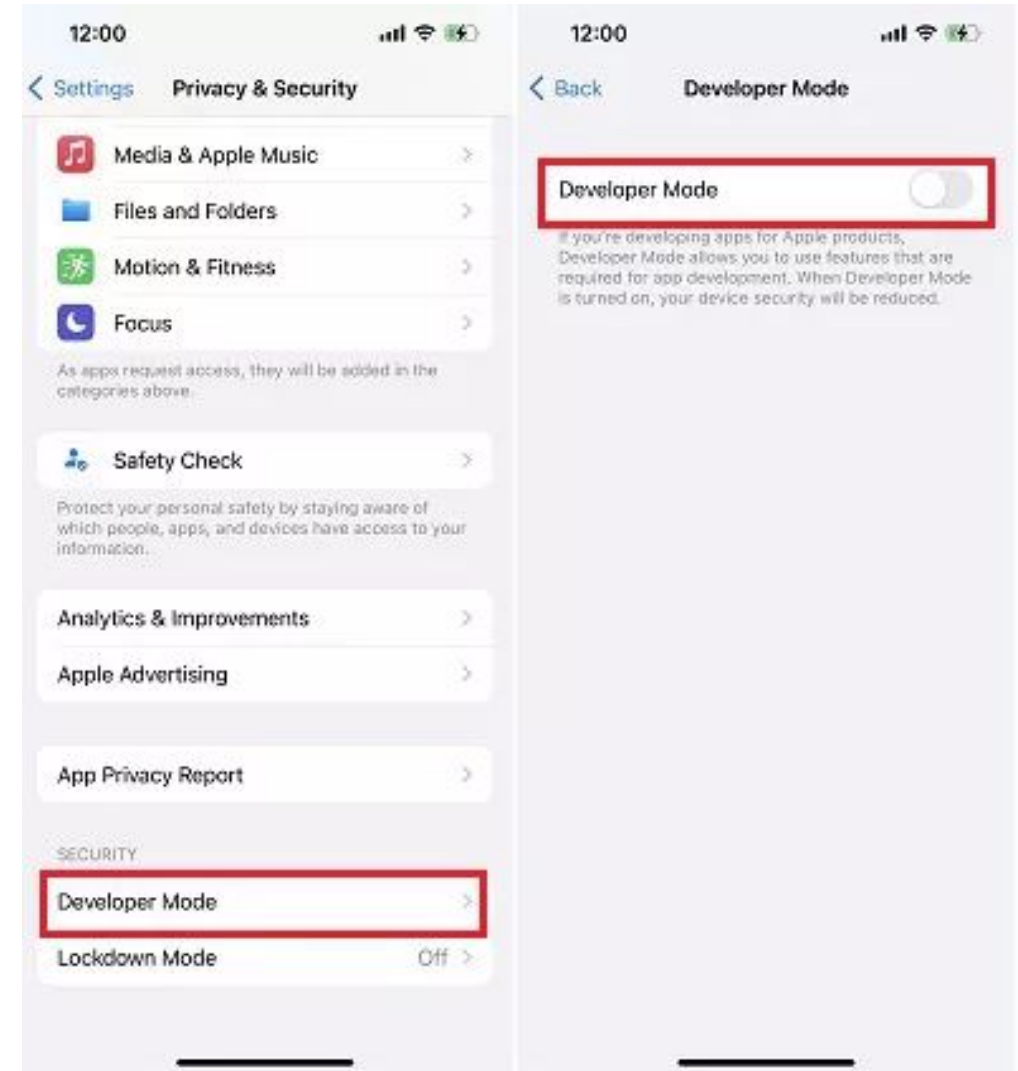
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 - 🔗 settings.gradle
 - 🔗 variables.gradle
 - > cypress
 - > dist
 - > ios
 - > node_modules
 - > public

The gradle.properties file contains the following code:

```
2
3 # IDE (e.g. Android Studio) users:
4 # Gradle settings configured through the IDE *will override*
5 # any settings specified in this file.
6
7 # For more details on how to configure your build environment visit
8 # http://www.gradle.org/docs/current/userguide/build_environment.html
9
10 # Specifies the JVM arguments used for the daemon process.
11 # The setting is particularly useful for tweaking memory settings.
12 org.gradle.jvmargs=-Xmx1536m
13
14 # When configured, Gradle will run in incubating parallel mode.
15 # This option should only be used with decoupled projects. More details, visit
16 # http://www.gradle.org/docs/current/userguide/multi_project_builds.html#sec:decoupled_projects
17 # org.gradle.parallel=true
18
19 # AndroidX package structure to make it clearer which packages are bundled with the
20 # Android operating system, and which are packaged with your app's APK
21 # https://developer.android.com/topic/libraries/support-library/androidx-rn
22 android.useAndroidX=true
23
24 org.gradle.java.home=C:\\Program Files\\Java\\jdk-17
25
```

Install into iOS

- ✓ Must have a MAC OS
- ✓ Must have xCode installed and configured in your MAC, also xCode CLI
 - ✓ After install, run this commands:
 - **xcode-select --install**
 - **sudo xcodebuild -license**, go to end of the text and write **agree**
 - ✓ Signing error – set apple ID: open xcode > Settings (o Preferences) > Accounts, then add your apple ID
 - ✓ Accept licenses and install additional components
- ✓ Developer mode on your phone must be enabled.
 - ✓ Go to Settings
 - ✓ Privacy and Security
 - ✓ Then, in developer mode, Enable "Developer Mode"
 - ✓ Finally, restart the phone.

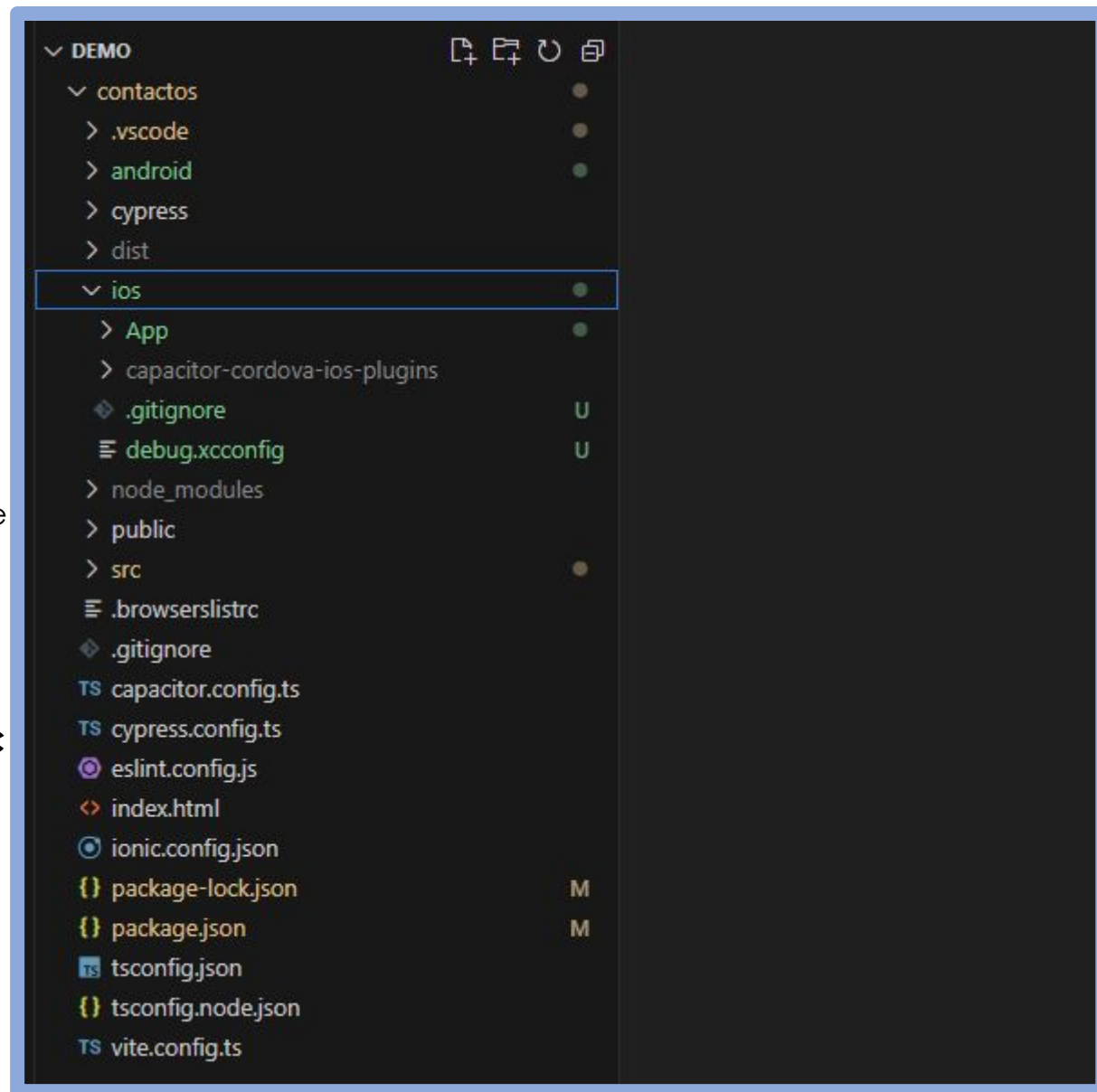


Install into iOS

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 - ✓ Accept licenses and install additional components
- ✓ Run this command in your visual studio project: **ionic cap run android**.

This command will run internally:

- ✓ ionic build: Builds the web app (React)
- ✓ npx cap copy ios: Copies the files to the /ios folder
- ✓ npx cap sync ios: Syncs plugins
- ✓ Open and run on xCode (emulator or device)



Fixing errors into iOS

- ✓ Sometimes, could be necessary fix this errors in MAC:
 - ✓ Signing error – set Apple ID: go to xCode, select **target App**, go to tab **Signing & Capabilities** and active **Automatically manage signing**. Then, select your apple ID into **TEAM**

Practice 01

Once the environments is installed, let's practice into ionic:

Based on the React app from the Challenge 01:

- It's not necessary to uninstall your PWA
- Let's migrate your app to ionic using only ionic components
- Verify that the app continues working, installing it in your phone.



Challenge 03

Create a new Task Manager app in ionic.

It should contain:

- ✓ States and effects, if it's necessary.
- ✓ Child and parent components – at least 3 components.
- ✓ View a task list
- ✓ Add new tasks
- ✓ Mark tasks as completed
- ✓ Delete tasks

